

ARITHMETIC MEAN

$$\frac{\sum x}{n} \rightarrow \text{ungrouped}$$

$$\frac{\sum fx}{\sum f} \rightarrow \text{grouped}$$

HARMONIC MEAN

$$\frac{n}{\sum (\frac{1}{x})} \rightarrow \text{ungrouped}$$

$$\frac{\sum f}{\sum (\frac{1}{x})} \rightarrow \text{grouped}$$

GEOMETRIC MEAN

$$\sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot x_4}$$

$$\text{antilog} \left(\frac{\sum \log x}{n} \right) \rightarrow \text{ungrouped}$$

$$\text{antilog} \left(\frac{\sum f \log x}{\sum f} \right) \rightarrow \text{grouped}$$

MODE

$$\hat{x} = L + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h$$

L = lower class boundary

f_m = frequency

f_1 = frequency preceding

f_2 = frequency following

h = class interval

MEDIAN

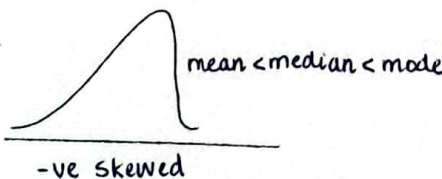
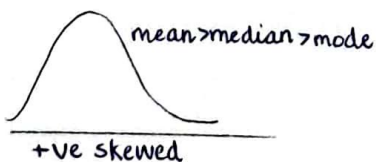
$$\tilde{x} = \left(\frac{n+1}{2} \right) \rightarrow \text{ungrouped}$$

$$\tilde{x} = L + \frac{h}{f} \left(\frac{n}{2} - CF \right) \rightarrow \text{grouped continuous}$$

L = lower class boundary

h = class interval

f = frequency



$$\text{Mode} = 3\text{median} - 2\text{mean}$$

QUANTILES (UNGROUPED)

$$Q_1 = \frac{1}{4}(n+1)$$

$$Q_3 = \frac{3}{4}(n+1)$$

$$D_k = \frac{k}{10}(n+1)$$

$$P_k = \frac{k}{100}(n+1)$$

QUANTILES (GROUPED)

$$Q_1 = L + \frac{h}{f} \left(\frac{n}{4} - CF \right)$$

$$Q_3 = L + \frac{h}{f} \left(\frac{3n}{4} - CF \right)$$

$$D_k = L + \frac{h}{f} \left(\frac{kn}{10} - CF \right)$$

L = lower limit

CF = preceding class f

STANDARD DEVIATION

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \rightarrow \text{ungrouped (population)}$$

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2}$$

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} \rightarrow \text{grouped}$$

$$\sigma = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2}$$

x = midpoint of class

VARIANCE

$$\text{variance} = \frac{\sum (x - \bar{x})^2}{n} \text{ (population)}$$

$$\text{variance} = \frac{\sum (x - \bar{x})^2}{n-1} \text{ (sample)}$$

COEFFICIENT OF SD

$$\frac{\sigma}{\bar{x}} = \frac{SD}{\text{mean}}$$

COEFFICIENT OF VARIANCE

$$\frac{SD}{\bar{x}} \times 100$$

QUANTILE DEVIATION

$$QD = \frac{1}{2}(Q_3 - Q_1) = SIQR$$

$$\text{Coefficient } QD = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

MEAN ABSOLUTE DEVIATION

$$MAD_{\bar{x}} = \frac{\sum |x - \bar{x}|}{n}$$

$$MAD_{\tilde{x}} = \frac{\sum |x - \tilde{x}|}{n}$$

$$MAD_{x^*} = \frac{\sum |x - x^*|}{n}$$

$$= \frac{\sum f|x - \bar{x}|}{\sum f}$$

$$= \frac{\sum f|x - \tilde{x}|}{\sum f}$$

$$= \frac{\sum f|x - x^*|}{\sum f}$$

ungrouped
grouped

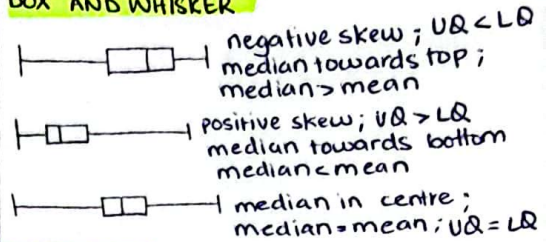
COEFFICIENT OF MAD

$$MAD_{\bar{x}} = \frac{MAD_{\tilde{x}}}{\bar{x}}$$

$$MAD_{\tilde{x}} = \frac{MAD_{x^*}}{\tilde{x}}$$

$$MAD_{x^*} = \frac{MAD_{x^*}}{x^*}$$

BOX AND WHISKER



OUTLIERS

$$\text{Lower Outliers} = Q_1 - (1.5 \times IQR)$$

$$\text{Higher Outliers} = Q_3 + (1.5 \times IQR)$$

EMPIRICAL FORMULA

68% data $\rightarrow$ mean - standard deviation
mean + standard deviation

95% data $\rightarrow$ mean - 2 standard deviation
mean + 2 standard deviation

99.7% data $\rightarrow$ mean - 3 standard deviation
mean + 3 standard deviation